

RODRIGO GOMES DE ARAÚJO

☎ +351 914 574 743 ✉ contact@rodrigoaraujo.pt 🌐 rodrigoaraujo.pt

EDUCATION

Master's in Computer Science

Faculdade de Ciências da Universidade do Porto (FCUP)

Sep 2025 – Present (Expected 2027)

Bachelor's in Computer Science and Engineering

Faculdade de Engenharia da Universidade do Porto (FEUP)

Sep 2022 – Jul 2025

PROJECTS

PoW Blockchain for Decentralized Auctions

Rust, Tokio, Dalek, libp2p, Tonic, Docker

- Developed a DHT overlay based on Kademlia to build the **decentralized network** and for peer discovery.
- Built a PoW blockchain using **Blake2b**, **Ed25519** signatures, and **Merkle trees** for transaction validation.
- Implemented block propagation with GossipSub, **chain synchronization**, block recovery through request-response, and a deterministic **longest-chain strategy** for fork resolution.
- Developed a **decentralized auction client** with consistent world-state updates across independent nodes.
- Created a **Docker-based adversarial testing environment** with miners, clients, honest nodes, and malicious peers to validate the implementation against the attack scenarios studied.

Embedded Distributed Synthesizer

C, POSIX, Embedded Systems, Real-Time Audio

- Developed a **distributed audio system** consisting of an Arduino controller, a Raspberry Pi audio engine, and an Android client for streaming and waveform visualization.
- Implemented a **modular low-latency audio engine** in C using miniaudio, supporting polyphony, waveform generation, modulation, and filtering.
- Designed a **serial communication protocol** that converts sensor inputs into validated packets processed as real-time audio transformations.
- Built a **lock-free concurrency model** using atomic shared state and ring buffers for event and packet handling.
- Implemented **UDP audio streaming** to Android using AudioTrack playback and a real-time oscilloscope visualizer.

ACHIEVEMENTS & AWARDS

2nd Place, IEEE RetroJam Game Development Competition (IEEE UP Student Branch)

2025

- Collaborated in a three-person team to design and implement a complete **2D game** within a **48-hour deadline** using Rust and Raylib. Composed the original soundtrack, designed in-game audio effects, and completed the final audio mixing and mastering.

Academic Merit Awards, Escola Secundária Carlos Amarante

2020–2022

- Recognized for outstanding academic performance in science and technology disciplines throughout high school.

World 4th Place, RoboCup Rescue Superteam (Leipzig, Germany)

2016

- Represented Portugal in an international autonomous robotics competition, developing navigation algorithms and autonomous systems for rescue scenarios.

National 1st Place, Robotics Championship (CoSpace Rescue)

2016

National 4th Place, Canguru Matemático Sem Fronteiras

2016

TECHNICAL SKILLS

Programming Languages: Rust, C/C++, Haskell, Python, Java, SQL

Frameworks & Tools: Git, Docker, UNIX, Tokio, POSIX, MPI, OpenMP, gRPC, libp2p, Dalek, Raylib, PostgreSQL